

MADL — Mobile Application Description Language

v0.1 · IDs are mandatory · closed sets are closed · screenshots are not specs

NAVIGATION — how the user moves

atlas	Complete deck/card/transition map. One per app.
stack	Push-down navigator. Back = pop.
tabset	Parallel persistent destinations.
drawer	Hidden navigator, revealed by gesture.
flow	Named linear deck sequence with exit.
entry	Default card shown on deck first open.
exit	Target on back / cancel / complete.

IDS

{app}.atlas	trk.atlas
{app}.nav.stk{n}	trk.nav.stk1
{app}.nav.tab{n}	trk.nav.tab1
{app}.nav.drw{n}	trk.nav.drw1
{app}.flow.{name}	trk.flow.onboard

TOPOLOGY

stack	→ drill-down, back-enabled
tabset	→ parallel top-level
drawer	→ secondary / settings
flow	→ linear task with end

● STRUCTURE ●

the addressable hierarchy

app	{app_id}
deck	{app}.d{n}
card	{deck}.c{n}
layer	{app}.l{n}
slot	{card}.s{n}

standard card names

default	empty
loading	loaded
error	editing
saving	success

INTERACTION — what the user does

trigger	Gesture or event initiating an action.
action	Operation bound to a trigger.
transition	Movement card→card or deck→deck.
guard	Condition gating a transition.
gesture	Physical trigger type. Closed set.

IDS

{card}.t{n}	trk.d2.c1.t1
{trigger}.a	trk.d2.c1.t1.a

GESTURES

tap	long-press	swipe-left	swipe-right	swipe-up
swipe-down	pinch	expand	scroll-end	back
system-event				

TRANSITIONS

push	pop	replace	modal-present	modal-dismiss
sheet-up	sheet-down	fade	none	

ELEMENTS — what populates cards

element	Base unit. Always typed.
field	Accepts user input.
label	Displays static or bound text.
control	Triggers an action.
media	Image, video, audio, map, chart.
list	Repeating bound collection.
overlay	Rendered above card surface.

IDS

{slot}.e{n}	trk.d2.c1.s2.e1
{card}.o{n}	trk.d2.c1.o1 (overlay)

OVERLAY SUB-TYPES

dialog	sheet	toast	tooltip	action-sheet
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KEY PROPERTIES

type	MUST on all elements
bound-to	MUST on list · SHOULD on media
visible-if	conditional render
modal	MUST on overlays
auto-dismiss	MUST on toast (seconds)

SERVICES — what the app talks to

service	Any external system.
store	Local on-device persistent data.
endpoint	Named operation on a service.
binding	Element ↔ data source link.
host	Device OS capabilities.
cloud	Remote platform service.
event	Async message from service.

IDS

{app}.svc.{name}	trk.svc.api
{app}.store.{name}	trk.store.db
{svc}.ep.{name}	trk.svc.api.ep.save
{app}.svc.host.{cap}	trk.svc.host.gps
{app}.svc.cloud.{name}	trk.svc.cloud.auth
{svc}.evt.{name}	trk.svc.api.evt.sync

HOST CAPABILITIES

camera	gps	biometrics	notifications	contacts
microphone	accelerometer	nfc	bluetooth	

SERVICE TYPES

api	→ REST / GraphQL / WebSocket
store	→ on-device · always available
host	→ device OS · native protocol
cloud	→ third-party platform SDK

IDs: lowercase · dot-separated · stable · mandatory